Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Andy Shiley	andyshi@udel.edu
Friend (optional)	Lena	mlta@udel.edu

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

My goal is to process eBird data and present it to a user, so that they can get unique insight on their birding experience, as well as ideas for which places would be optimal for birdwatching.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

The eBird website already has some built-in data representation that users can see, such as total species seen at a given place, total species seen, and total number of checklist. However, it does not have data on individual birds seen or quantities of birds seen. Thus, while an area might have a large amount of species richness (different species), it might not have a large number of birds. The latter might provide unique insight on where one might want to birdwatch, based on their previous experience.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

This application would:

- Prompt a user to input an eBird .csv file.
- Provide a list of all birding locations someone has visited
- Provide total bird count, species count, and number of each species for each place
- Provide total list of all birds you have seen, the number of that species you have seen, and the places you have seen that species
- Provide a summary of someone's stats in a ‘Spotify Wrapped’ style summary

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

Use an exported file of someone's eBird data and process it, to present it to the user in an interesting way, providing insight into the users birding habits and how they should continue.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will need to put an eBird data export in .csv format in the same folder as the program,and type the file name into an input box.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The computer will turn the data into a list of dataclasses, sum the numbers of species in the file, sum the number of birds in the file, and list the species in the file as well as the places in the file.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will receive their formatted data, which will provide insight into how they birdwatch, and where they should birdwatch to see a specific species or a number of species.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

- Bird Species
- Places
- Bird Sightings

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

These are based on the details the raw eBird data export has:

- Bird Sightings Needed Details:
    - Common Name
    - Scientific Name
    - Count
    - State/Province
    - County
    - Location
    - Time
    - Date
- Bird Sightings Ignored Details:
    - ML Catalog Numbers
    - Checklist Comments
    - Observation Details
    - Breeding Code
    - Number of Observers
    - Distance Traveled (km)
    - Area Covered (ha)
    - All Obs Reported
    - Duration (min)
    - Protocol
    - Latitude/Longitude
    - Location ID
    - Taxonomic Order
    - Submission ID

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The most difficult part of this project will be processing the raw data. EBird data is super robust and there are many different components of it that need to be abstracted and organized. Additionally, formatting drafter will be challenging.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I am not using an LLM for code. I will work on this project on Wednesday or Thursday Night (5-11 pm).

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

I want to build something that is useful and fun for my hobby, that might help me find out how to improve at my hobby. It could also be used by my friends and other people in the birdwatching community.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a website where you can explore your eBird data, and use it to help you decide where to bird in the future. How do you decide where you should go birding, and what is some eBird data you would like to see explored?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

- *They birdwatch based on location/convenience*
- *They sometimes chase a rarity if it is likely in x location at y time*
- *They think it would be cool to use all ebird data and compare it with you own, then have the app recommend places to birds to see new birds you haven't seen before.*
- *They would like total species and individual counts at different locations*
- *They would like a list or map of all the places they've gone birdwatching.*

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

Accessing all eBird data is not possible because that data is not public, and it would be too much to run locally. It would take a very long time to process all that data without more advanced filtering techniques. What could be a simpler alternative?
They said:

- Being able to filter based on you own data which species are likely where, depending on time of year
- Comparing data to a subset of known information for a smaller region, such as which birds are likely in delaware

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

I think it has given me good ideas for where I could expand the project to, in order to make it more robust and helpful. Depending on how many of the project requirements I have met when I execute my initial ideas, I may expand it to incorporate some ideas from this conversation.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I will use the BlockPy textbook as a reference for working code. I will also use the discord and TA office hours if I cannot figure out how to proceed.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

- Rubber Ducky debugging:
  - Explaining my code out loud
  - FInding a gap in my logic
- Wolf-Fence Debugging:
  - Splitting my code in half
  - Seeing if the error persists in which half
  - Splitting that half again
  - Repeating the procedure until I find the line that isn't functioning as intended

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use an LLM.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.

Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

The concepts from extracting data from files will be useful to this project. I expect to use primarily loops and nested data (lists of dataclasses with lists in them) in this project. I want to improve my ability to “think” in code, increasing my fluency in problem solving natively in python. I also want to improve my drafter skills.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

Extracting data from the files will be easy, as it just involves basic loop appending to a list. However, it will be much harder to format that data in logical lists of dataclasses, as that involves complex construction and nested data.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

The most time and effort will be spent on the backend data processing formatting,

while the actual presentation of the data in drafter should be relatively easy. I will implement turning a datafile into some list of dataclasses with only relevant information first, so that I can construct more specific data classes using that data. I estimate the time to complete the backend portion will be around 4 hours. The frontend programming should only take 1-2 hours.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Other classes and assignments will interfere with my work. I will minimize them by strategically allocating time for all of my class assignments. I don't have any more questions about requirements.

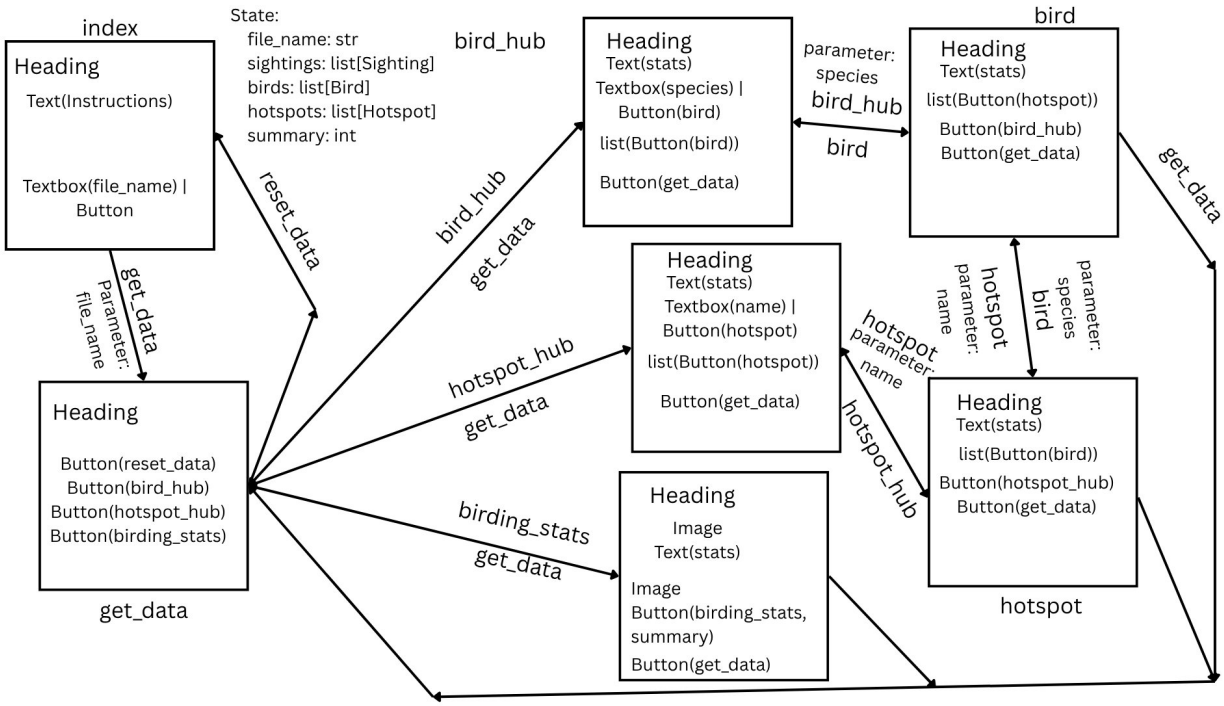
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


index
Heading
Text(Instructions)
Textbox(file_name) | Button
State:
file_name: str
sightings: list[Sighting]
birds: list[Bird]
hotspots: list[Hotspot]
summary: int
reset_data
get_data
Parameter: file_name
Heading
Button(reset_data)
Button(bird_hub)
Button(hotspot_hub)
Button(birding_stats)
get_data
bird_hub
get_data
bird_hub
Heading
Text(stats)
Textbox(species) | Button(bird)
list(Button(bird))
Button(get_data)
parameter: species
bird_hub
bird
bird
Heading
Text(stats)
list(Button(hotspot))
Button(bird_hub)
Button(get_data)
get_data
hotspot_hub
get_data
Heading
Text(stats)
Textbox(name) | Button(hotspot)
list(Button(hotspot))
Button(get_data)
hotspot
parameter: name
hotspot
parameter: name
hotspot
parameter: species
bird
hotspot_hub
Heading
Text(stats)
list(Button(bird))
Button(hotspot_hub)
Button(get_data)
hotspot
birding_stats
get_data
Heading
Image
Text(stats)
Image
Button(birding_stats, summary)
Button(get_data)

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**The site will process eBird data and allow the user to inspect specific aspects of it such as sightings/counts of a specific bird, or at a specific location. It will also create a spotify wrapped style "summary" of the users birding statistics.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	file_name: str The name of the user's ebird data file (csv). This will be filled by the typed name of a user's file.Reset to '' by reset_data route. sightings: list[Sighting] A list of Sighting dataclasses. This will be constructed in the get_data route route (from [] to a full list), based on the user's data. Reset to [] by reset_data route. birds: list[Bird] A list of Bird dataclasses. This will be constructed in the get_data route route (from [] to a full list), based on the sightings attribute. Reset to [] by reset_data route. hotspots: list[Hotspot] A list of Hotspot dataclasses. This will be constructed in the get_data route route (from [] to a full list), based on the sightings attribute. Reset to [] by reset_data route. summary: int An integer representing which page of the summary should be displayed. This will be increased by one every time a page is displayed, and reset to 0 on the

	final summary page. Reset to 0 by reset_data route.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Sighting: species: str count: str state: str county: str name: str These will not be modified, only created by get_data and reset by reset_data. Bird: species: str count: int hotspots_bird: list[str] Hotspot: name: str total: int birds_local: list[str] location: str Both Bird and Hotspot will be updated during get_data. If a sighting has a bird or hotspot name that already has an existing dataclass instance, the count/total integer, and birds_local/hotspots_bird list[str] will be updated to reflect this. Also reset by reset_data.
Routes (minimum 7) The names of your route functions	index get_data reset_data bird_hub hotspot_hub birding_stats

	bird hotspot
Pages (minimum 5) A brief description of the pages in your site	index ● Lets the user input the name of their file to start the site get_data ● Home page, has buttons to bird and hotspot hubs, as well as the summary stats page. ● Has a button to reset the state dataclass and go back to index bird_hub ● Stats about how many birds you have seen. ● Textbox to search for a species of bird ● List of buttons that take you to bird pages with their corresponding species. ● Button to go back to get_data hotspot_hub ● Stats about how many hotspots you have visited. ● Textbox to search for a name of hotspot ● List of buttons that take you to hotspot pages with their corresponding name. ● Button to go back to get_data birding_stats ● A stack of pages that displays an image of a bird, a statistic calculated based on their data, and a button linking to the next birding stats page. ● On the last page, a button to get_data bird ● Species name ● Text with number of this species seen ● Button for all hotspots the bird was sighted at, that link to hotspot pages. ● Button to bird_hub, get_data hotspot ● Hotspot name, location, number of birds seen there

	• Button for all birds seen at hotspot, that link to bird pages. • Button to hotspot_hub, get_data
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	TextBox 1: Allows the user to type in the name of their eBird data export file. TextBox 2: Allows the user to type in the name of a bird species to bring up that bird page. TextBox 3: Allows the user to type in the name of a hotspot to bring up that hotspot page.
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	• Statements to check the value of summary, which will be used to display different summary data pages (5 if statements) • If statements to check if the typed name or species of a bird/hotspot is on the list of birds/hotspots (2 if statements) • Statement to check if a sighting was reported as x in raw data (meaning count unknown) -> will replace it with 1, minimum conservative estimate • Statement to check if hotspot/bird species/name already has a dataclass, will update the data in that class if preexisting, or create a new instance of Bird/Hotspot dataclasses
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	• For loops to add buttons to the bird_hub/hotspot_hub pages. THis is meaningful to let users access related information. • For loops for summary statistics. Some summary stats (like total birds seen, or most birds seen at a hotspot) will use the sum and max patterns to calculate the relevant statistics. These will be displayed to the user, which is a key part of the site. (2 loops)

	● For loops using search patterns to bring up bird/hotspot pages. Meaningful for user convenience, not having to scroll down an entire list (2 loops). ● For loops to check if a Hotspot/Bird dataclass with a matching name/species already exists. Meaningful to accurately calculate statistics for birds and hotspots (2 loops). ● For loop to parse raw data and create sighting dataclasses. Meaningful because it is the way the site will access eBird data.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

No LLM use.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.